DIGITAL TWIN CHALLENGE

Personalized Glucose Risk Prediction

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Why Predict Glucose Risk?

The problem

Glucose levels in people with Type 1 diabetes can fluctuate throughout the day. A single reading may not capture how glucose is changing or whether a high-glucose event may be approaching.

Our objective

Explore whether the previous **2 hours of physiological data** can help predict the onset of a qualifying high-glucose event within the **next 30 minutes**.

Understanding the Dataset

Dataset: HUPA-UCM Diabetes Dataset

Data used

- Dynamic features: Glucose, heart rate, calories, steps, basal insulin, bolus insulin, and carbohydrate intake.
- Static features: Age, gender, HbA1c, diabetes duration, body measurements, BMI, and treatment.

INSIDE THE DATASET

Dynamic Time-Series Data

time	glucose	calories	heart_rate	steps	basal_rate	bolus_volume_delivered	carb_input
2018-06-26T23:20:00	163.333333	5.59674	70.731707	0.0	0.075	0.0	0.0
2018-06-26T23:25:00	160.666667	5.59674	68.875000	0.0	0.075	0.0	0.0
2018-06-26T23:30:00	158.000000	5.48700	71.621622	0.0	0.075	0.0	0.0
2018-06-26T23:35:00	159.000000	5.70648	71.692308	0.0	0.075	0.0	0.0
2018-06-26T23:40:00	160.000000	5.48700	70.346154	0.0	0.075	0.0	0.0
2018-06-26T23:45:00	161.000000	5.48700	71.437500	0.0	0.075	0.0	0.0

Static Patient Metadata

gender	hba1c_percent	age_years	diabetes_duration_years	weight_kg	height_cm	treatment	bmi
Female	8.2	56.3	15.5	59.0	161	CSII	22.76
Male	7.1	48.6	36.5	82.4	186	CSII	23.82
Male	7.3	43.4	12.5	62.0	182	CSII	18.72
Male	7.8	41.2	8.5	88.0	180	CSII	27.16
Female	6.9	41.9	39.5	58.5	161	CSII	22.57

How Does Our Model Predict Risk?

1. Input window

The previous 25 readings, sampled every 5 minutes, across seven dynamic features.

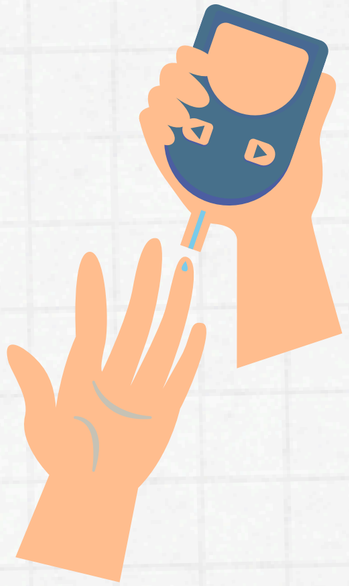
2. Event definition

A qualifying event requires glucose above 180 mg/dL for six consecutive readings.

3. Prediction target

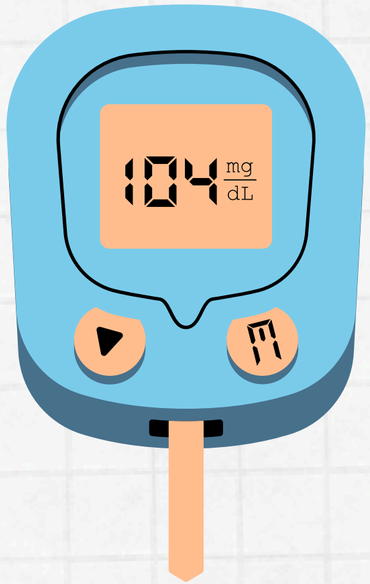
Predict whether a qualifying high-glucose event will begin within the next 30 minutes.

Understanding the Prediction Window



01 — Input History

25 readings · 5-minute intervals · 2 hours
The model uses seven dynamic features from the previous 25 timestamps.



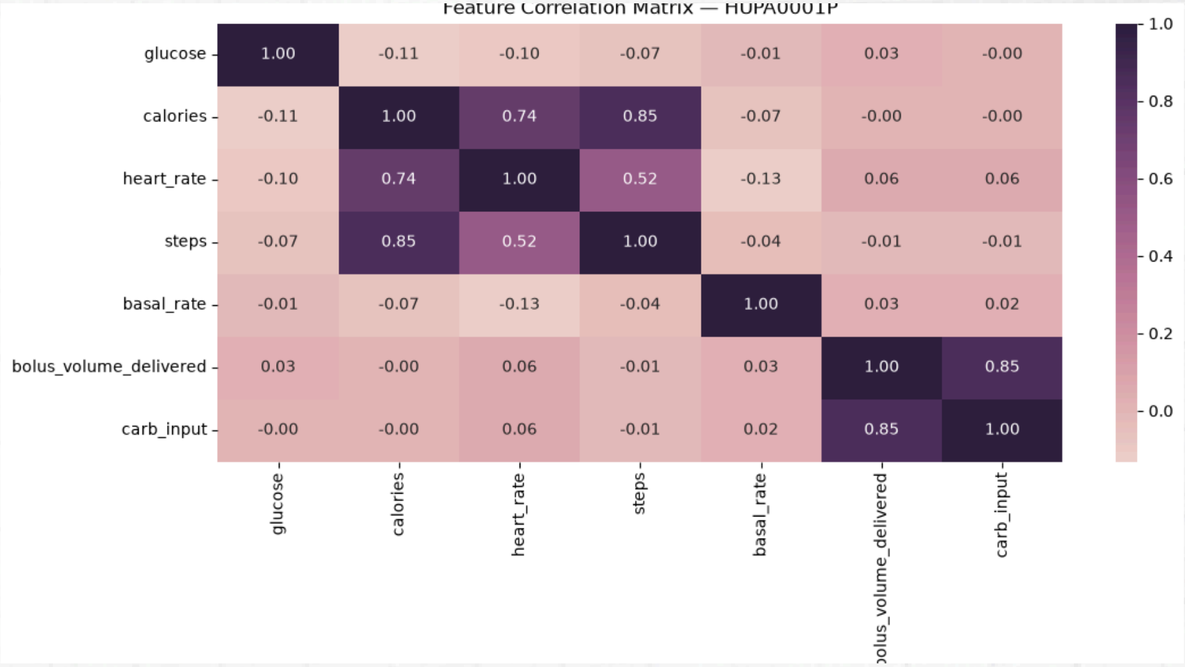
02 — High-Glucose Event

Above 180 mg/dL · 6 consecutive readings
All six readings must exceed 180 mg/dL, representing approximately 30 minutes of sustained high glucose.

03 — Prediction Horizon

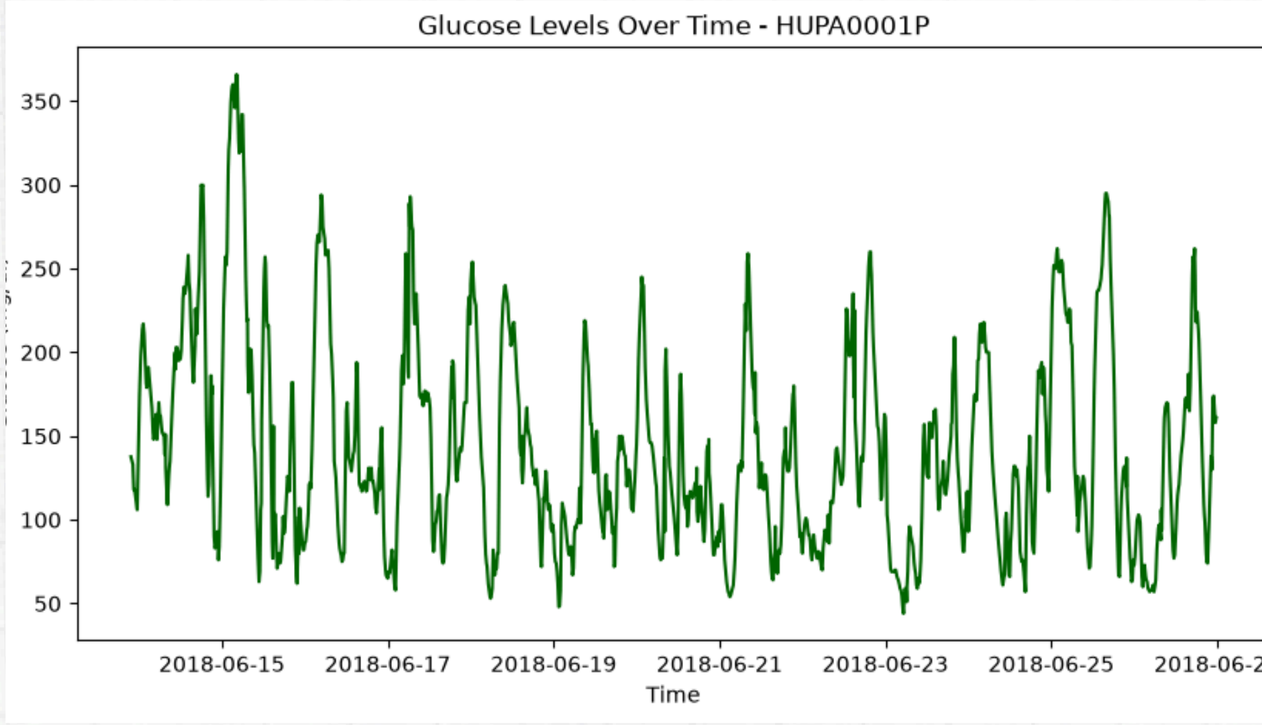
The next 30 minutes
Predict whether a qualifying event will begin within this future window.

Exploring Glucose Patterns



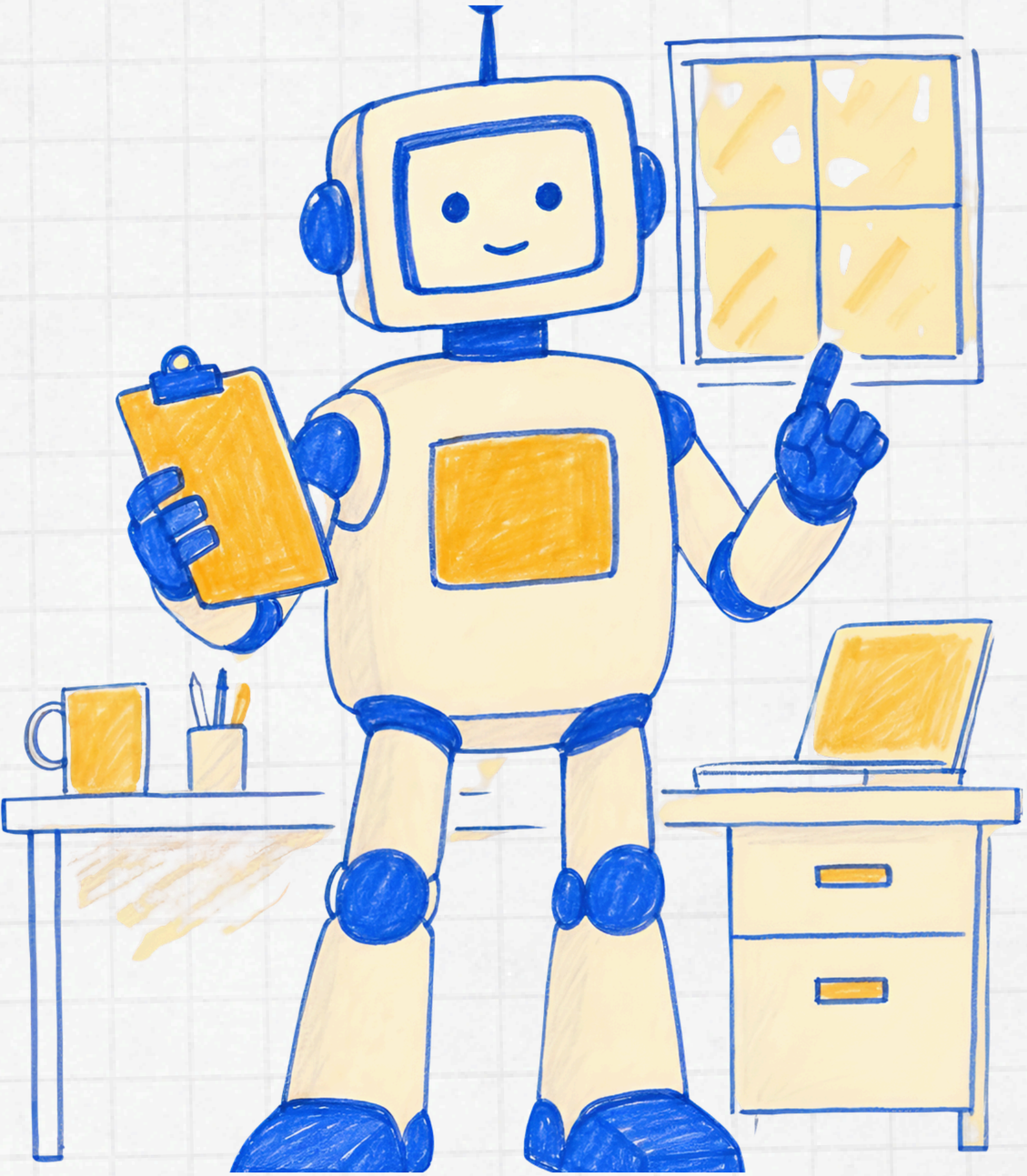
1. Correlation Matrix

What it shows: Linear relationships between numerical features in the dataset.



2. Glucose Over Time

What it shows: How glucose changes over time and when it crosses the 180 mg/dL threshold.



Preventing Data Leakage

Patient-Wise Data Splitting

25 participants → Three independent groups

- Training: 17 patients — 60,402 windows
- Validation: 4 patients — 32,116 windows
- Testing: 4 patients — 9,730 windows

Why This Matters

Overlapping time-series windows from the same patient can be highly similar. Randomly splitting windows could allow patient-specific patterns to appear in both training and testing data.

By assigning entire patients to separate splits, we evaluated the model on participants it had not seen during training.

Preprocessing Without Leakage

- Dynamic and static scalers were fitted using training data only.
- The static categorical encoder was fitted using training data only.
- The decision threshold was selected on validation data, not the test set.

Key takeaway: The held-out test set measures performance on unseen participants, providing a more meaningful generalization check.

25 patients

Train
17
60,402 windows

Validation
4
32,116 windows

Test
4
9,730 windows

Each patient belongs to exactly one .

Model Experiments

Logistic Regression

Baseline
A simple reference model for binary classification.

CNN

Convolutional
Learns local patterns in sequential glucose data.

LSTM

Recurrent
Learns temporal dependencies across readings.

GRU

Recurrent
Learns sequential patterns using gated recurrent units

Inside the Digital Twin: Model Architecture

Dynamic history
25 time steps × 7 features

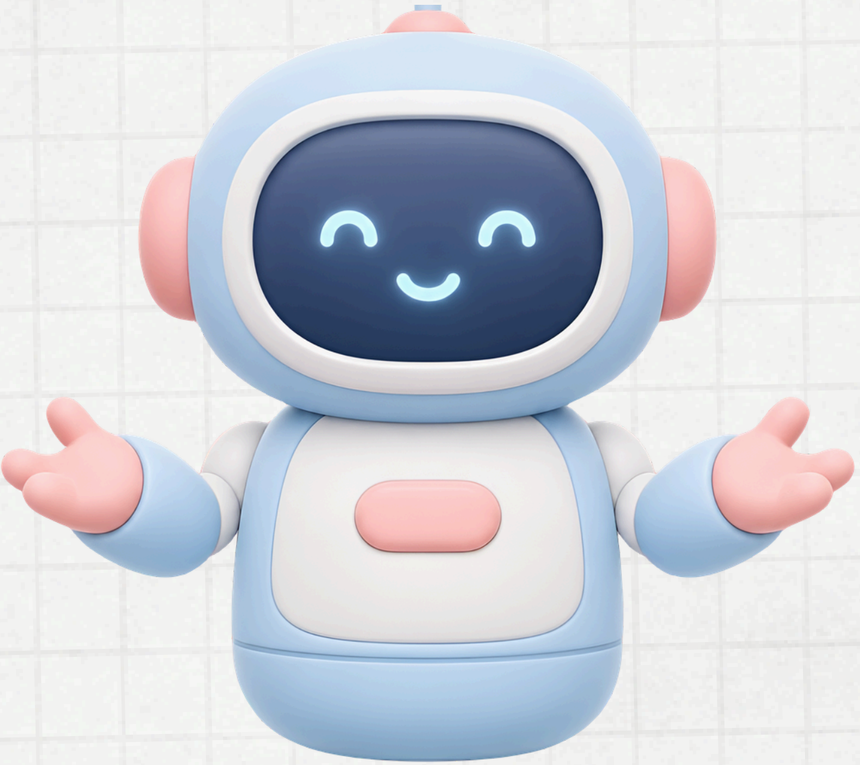
GRU sequence encoder
Learns temporal patterns

Patient metadata
10 encoded/scaled features

GRU + Static → Risk Prediction

Model	ROC-AUC	PR-AUC	F1-score
Logistic Regression	0.898	0.440	0.478
CNN	0.845	0.342	0.393
CNN+Engineered	0.877	0.394	0.412
LSTM	0.914	0.493	0.515
GRU + Static	0.911	0.491	0.523
GRU — Dynamic only	0.928	0.549	0.583

KEY TAKEAWAYS FROM MODEL EXPERIMENTS



01 — GRU Captured Temporal Patterns

The dynamic-only GRU achieved the strongest validation performance, with a ROC-AUC of 0.928, PR-AUC of 0.549, and F1-score of 0.583.

02 — Static Features Did Not Always Improve Performance

Adding patient metadata did not outperform the dynamic-only GRU on every metric. This highlighted the importance of testing whether additional features genuinely improve predictive performance.

03 — Class Imbalance Required Careful Evaluation

Positive windows represented only about 5.38% of the dataset. Therefore, we evaluated PR-AUC, precision, recall, and F1-score alongside ROC-AUC rather than relying on accuracy alone.

04 — Model Selection and Deployment Had Different Goals

Although the dynamic-only GRU performed best on validation, we deployed the GRU + static model to explore the digital-twin concept using both physiological history and patient characteristics.

Evaluating Our Digital Twin

Final GRU + static model · Held-out test set

ROC-AUC	PR-AUC
0.856	0.331

Recall	Precision
67.7%	26.6%

Metrics are from the held-out test set, using the decision threshold selected on validation data.

The model identified about 68% of positive test windows, but many positive predictions were false alarms.

Final Evaluation — Held-Out Test Set

GRU + Static Model

ROC-AUC	PR-AUC
0.856	0.331

Recall	Precision
67.7%	26.6%

What do the results tell us?

- Detected 458 of 677 positive test windows.
- Missed 219 positive windows.
- 1,263 false positives — many alerts were false alarms.

Key takeaway: The model detected 67.7% of positive windows, but low precision means many positive predictions were false alarms.

Threshold selected on validation data · Test set: 4 held-out participants · Experimental prototype, not a clinically validated medical tool.

Where Did Our Predictions Go Right or Wrong?

False Positives (1,263)

The model predicted a high-glucose event, but no qualifying event began within the next 30 minutes. These false alarms reduce precision.

True Positives (458)

The model correctly predicted the onset of a qualifying high-glucose event within the prediction window.

False Negatives (219)

A qualifying event began within the next 30 minutes, but the model failed to predict it. These are missed events.

True Negatives (7,790)

The model correctly predicted that no qualifying event would begin within the next 30 minutes.

Limitations & Future Improvements

What Can Be Improved?

Limited generalization

The test set contains only four held-out participants, so results may not generalize to other people.

False alarms

Low precision means many positive predictions did not correspond to a qualifying event onset within the forecast window.

Not clinically validated

This is an experimental proof of concept, not a medical device or a substitute for glucose monitoring or clinical advice.

Future work

Evaluate more participants, improve precision, and validate the model prospectively before considering real-world use.

Live Demo — Meet Your Digital Twin

From historical patient data to a deployed risk prediction

01 — Access the Application

Open the deployed web application in a browser.

02 — Provide Patient Inputs

Upload the time-series CSV and enter the required patient characteristics.

03 — Run the Model

Generate a prediction using the deployed GRU + static model.

04 — Interpret the Output

View the estimated risk probability, prediction, decision threshold, and glucose history.

Deployment Architecture + Live Demo

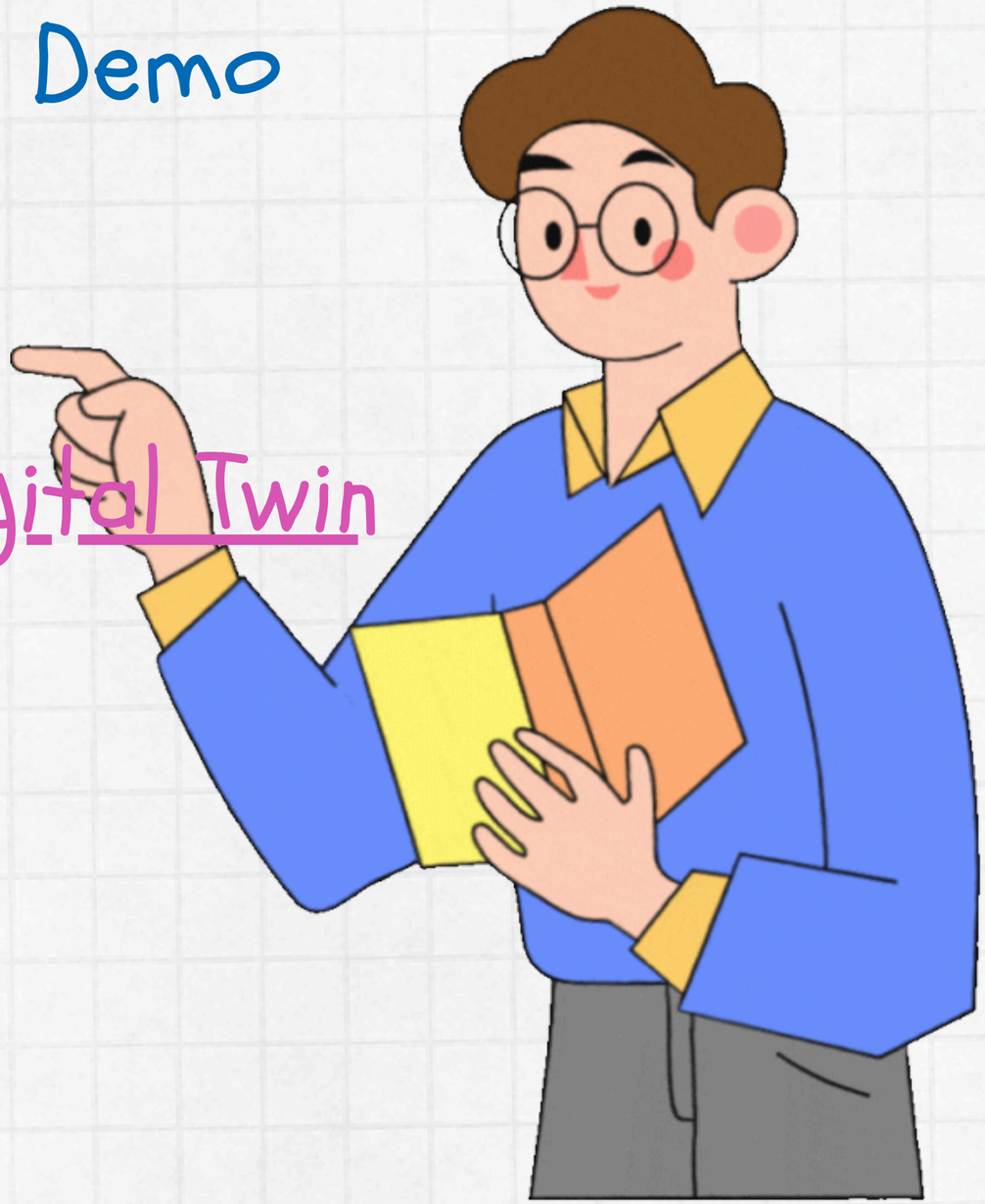
React Frontend — Vercel
CSV upload + patient profile

FastAPI Backend — Render
Input validation + preprocessing

GRU + Static Model
Risk probability + threshold-based prediction

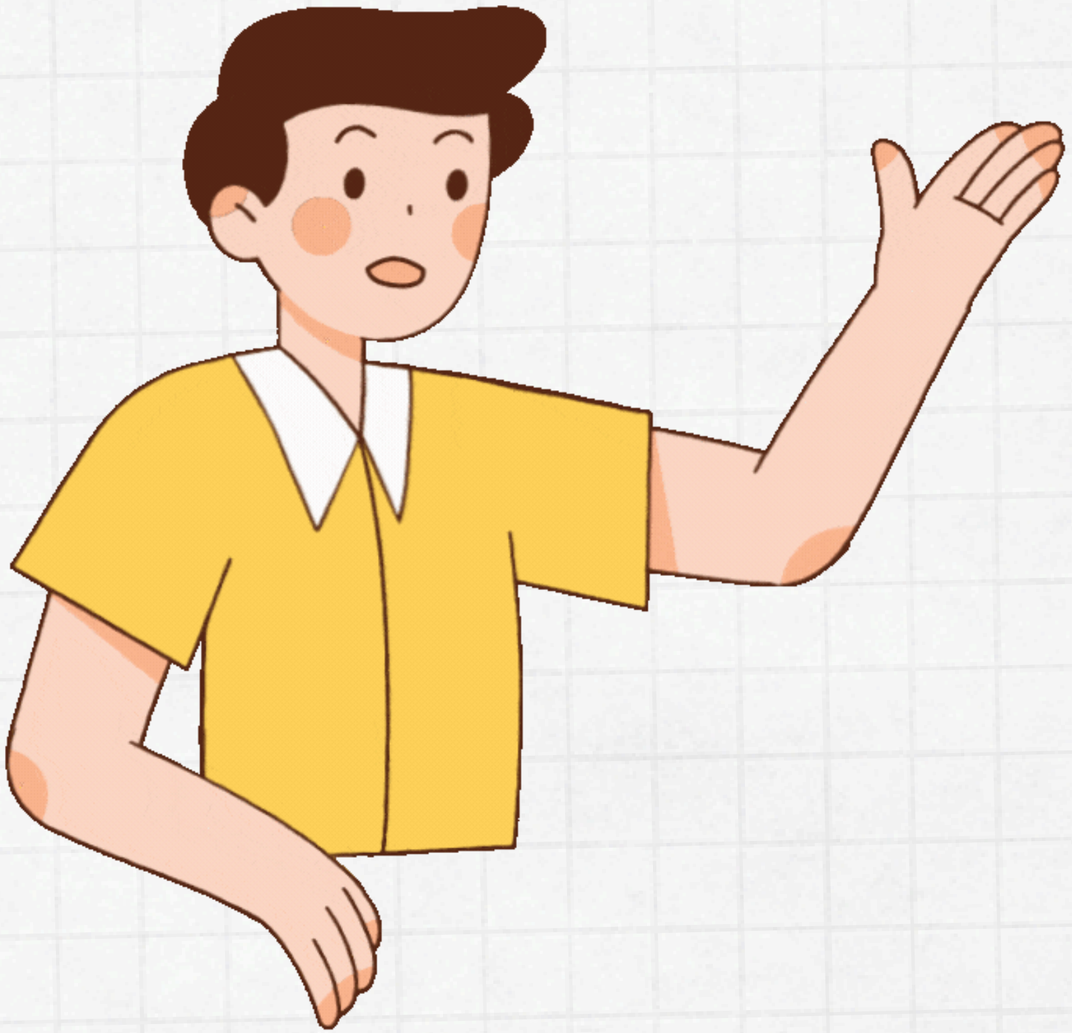
Results Dashboard
Prediction, glucose history and visualizations

Live demo: Meet Your Digital Twin



Defining the Target Event

Turning continuous glucose readings into a prediction task



180 mg/dL
Glucose must be strictly above this threshold.

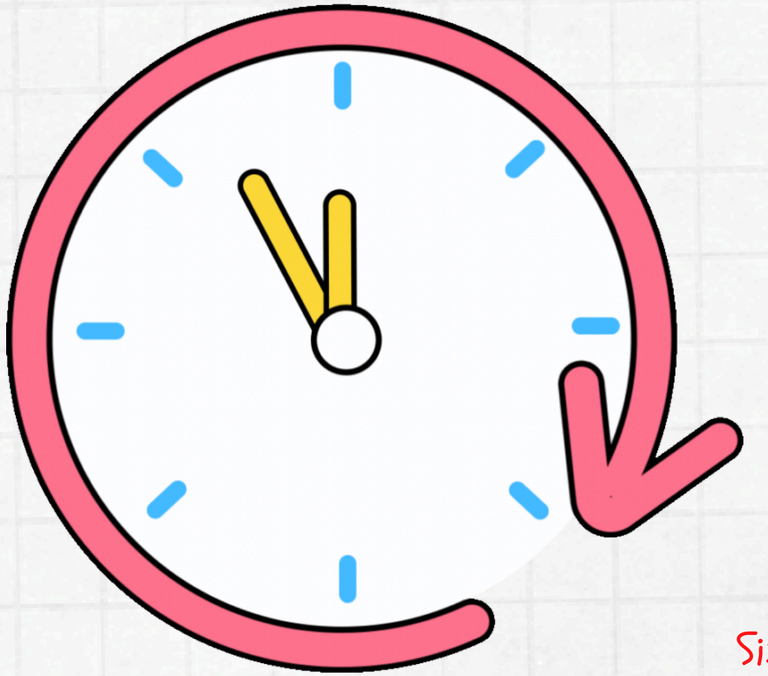
6 readings
Six consecutive readings above the threshold.

Prediction framework

2 HOURS
HISTORICAL DATA

→

30 MINUTES
PREDICTION HORIZON



- Sampling interval: 5 minutes
- Positive label: A qualifying event begins within the next 30 minutes.
- Exclusion: Prediction windows already inside an active qualifying event.

Six consecutive readings span 25 minutes between the first and sixth timestamps, representing approximately 30 minutes of sustained elevated glucose.

From Probability to Prediction

How the model converts its output into a binary classification

Selected probability threshold

0.7496

Approximately 74.96%

PROBABILITY $\geq$ 0.7496

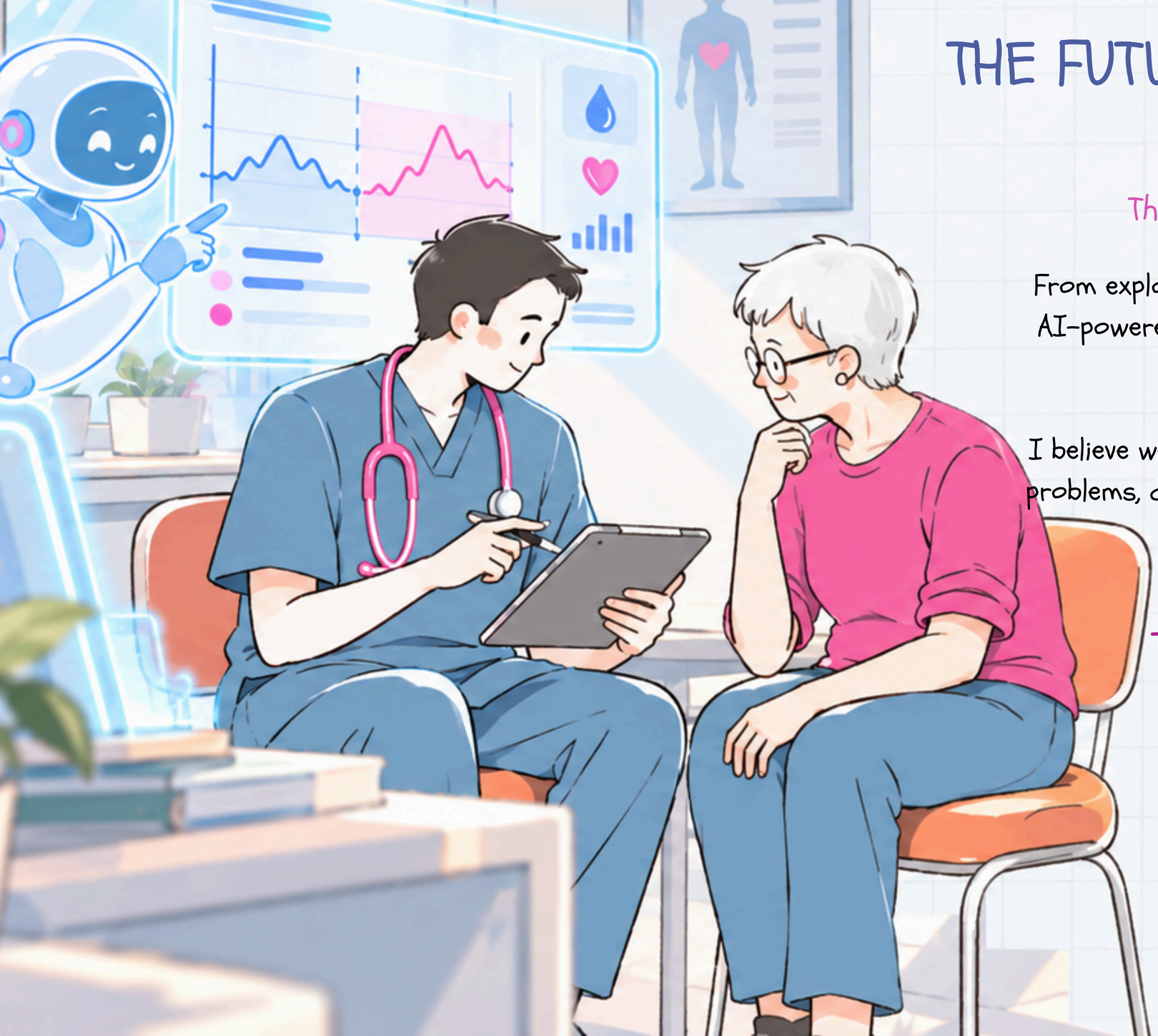
PREDICT A QUALIFYING EVENT.

PROBABILITY $<$ 0.7496

PREDICT NO QUALIFYING EVENT.

How was the threshold selected?

- Evaluated candidate thresholds on the validation set.
- Selected the threshold that maximized the F1-score.
- Kept the selected threshold fixed for final test evaluation.



THE FUTURE IS BEING BUILT TODAY

Thank You, Digital Twin Challenge 2026!

From exploring real-world glucose data to building and deploying an AI-powered prototype, this project has been a journey of learning, experimentation, and innovation.

I believe we're entering an era where AI can help us explore difficult problems, discover new possibilities, and develop solutions that once seemed out of reach.

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This isn't just about AI taking over. It's about what we, as humans, can create with AI.

Thank You! 💙